

Figure 1. FruitCropXL model cards

A modular functional–structural fruit-crop modelling platform

1. Model identity



FruitCropXL

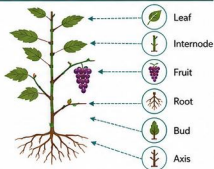
Modular functional–
structural fruit-crop model

Implemented in
GroIMP / XL

Current focus:
apple and grapevine
systems

Scale: organ →
shoot/axis → plant → field

2. Biological structure



- Represents connected 3D plant structure
- Structural levels: organ, metamer, axis, plant
- Topology and geometry drive local microclimate and resource dynamics

3. Inputs and configuration



model.options.*.json

species, module switches,
architecture mode,
simulation window, outputs,
management/planners



plant.parameters.*.json

organ traits, allometry,
photosynthesis, hydraulics,
carbon and fruit
parameters



initial.conditions.*.json

starting biomass, organ
state, fruit state,
root/wood/leaf
initial values



Weather — radiation,
temperature,
humidity/VPD,
rainfall/irrigation



Orchard design —
spacing, row
orientation, training system



Plant architecture —
CSV/generator outputs,
bud maps, root structure

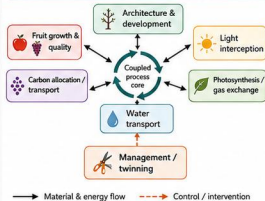


Management —
pruning, thinning,
shading, water restriction,
twinning

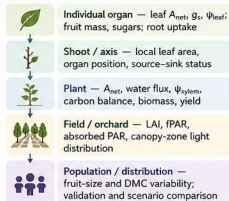


Support data — phenology,
shading
factors, organ-size profiles,
soil data

4. Core process modules



5. Output layers and evaluation



6. Intended use and current boundaries

Intended use

- Scientific analysis
- Scenario simulation
- Virtual experiments
- Digital-twin development
- Decision-support prototyping
- Platform extension

Current boundaries

- Scenario-specific parameterisation needed
- Some dynamic processes are still being refined
- Validation confidence depends on available data
- Best used within tested species, systems and module settings



Configuration
+ inputs



3D plant/root
structure



coupled
processes



multi-scale
outputs



use cases